Ayush Bansal

(+1) 341.529.6471 | ayushb6@illinois.edu | ayush268

RESEARCH INTERESTS

Quantum Systems · Distributed Systems & Networking · Operating Systems · Cloud Computing

EDUCATION

University of Illinois Urbana-Champaign (UIUC) 2024 – Present
Ph.D. in Computer Science Urbana, Illinois

- Working on **Quantum-Centric Supercomputing (QCSC)**, advised by Prof. Tianyin Xu.

Indian Institute of Technology Kanpur (IITK) 2016 – 2020
B.Tech. in Computer Science and Engineering (CGPA: 9.0/10.0) Kanpur, India

- Specialized in Systems (kernel, distributed systems, networking) and Programming Languages.

PROJECTS

Understanding and Fixing Bottlenecks in QCSC Pipelines Feb 2026 – Present
Under Prof. Tianyin Xu and Dr. Apoorve Mohan UIUC

- Build efficient **fault-tolerant QCSC pipelines** for stable fidelity with minimal quantum-classical resource usage.
- Built a **QEM profiler** exposing multiple optimization dimensions in the Quantum Error Mitigation pipeline; designing an end-to-end **workflow manager** for quantum workloads over a shared HPC stack (CPUs, GPUs, QPUs).

My way or the PHY way: Every Packet deserves the right MCS Jan 2025 – July 2025
Under Prof. Brighten Godfrey and Prof. Matthew Caesar UIUC

- Designed an **application-aware rate-control mechanism** optimizing per-packet Modulation and Coding Schemes (MCS) based on traffic intent (latency vs throughput).
- Cut tail latency for real-time traffic by eliminating mandatory probing (10–15% overhead) used in algorithms like Linux Minstrel, exposing application semantics to the PHY/MAC layers.

Olympus: Semantic query framework for satellite imagery Jan 2025 – Dec 2025
Under Prof. Indranil Gupta UIUC

- Engineered a **real-time semantic query framework** for satellite imagery, shifting from image-centric to representation-centric retrieval.
- Integrated geospatial foundation models (SatlasNet) for onboard embeddings, achieving a **100x downlink reduction** and enabling direct parallel querying without image access.

EXPERIENCE

Siebel School of Computing and Data Science, UIUC Aug 2024 – Present
Graduate Research Assistant Urbana, Illinois

- Research on **Quantum-Centric Supercomputing**: co-designing quantum-classical systems that schedule quantum workloads across CPUs, GPUs, and QPUs to maximize fidelity at least resource cost.

NimbleEdge Mar 2021 – Feb 2024
Co-Founder and Chief Technology Officer Bangalore, India

- Raised a **$3.3M pre-seed** (Neotribe Ventures, Sistema Asia Capital) and built the entire technology org, migrating compute from the cloud to edge devices.
- Onboarded **OYO, Dream11 & Glance (InMobi)** as clients (each 100M+ MAUs); built a P2P SDK and aggregation infrastructure scaling to hundreds of millions of devices.

Tower Research Capital Aug 2020 – Mar 2021
Low Latency Software Developer Gurgaon, India

- Owned the firm-wide **Trade Reconciliation System** across all markets, ensuring correct functioning of the **trading platform** and handling **network/FPGA** errors; extended it to Taiwan Futures and crypto exchanges (FTX, Huobi).
- Built low-latency C++/Python **simulation software** to test trading strategies against custom in-house exchanges on historical market data.

Core Engineering, Tower Research Capital May 2019 – July 2019
Software Developer Intern Gurgaon, India

- Containerized and automated the Trade Reconciliation setup as **microservices** on bare-metal Kubernetes.
- Built low-latency C++ software inducing deterministic mutations on the network stream to simulate **trading-platform** and **FPGA** failure scenarios; earned a **Pre-Placement Offer** out of ∼35 interns.

InterviewVector June 2020 – Aug 2020
Co-Founder and Chief Technology Officer Gurgaon, India

- Created the **Interview-as-a-Service** category; signed **10+ clients** (BharatPe, MindTickle) and bootstrapped to **$170K ARR in 4 months**.

Eightfold.ai Apr 2020 – June 2020
Software Developer Intern Noida, India

- Built the Alert Management System and set up BDD testing with Selenium, cutting test-suite runtime from **62 to <10 minutes (84%)**.

Google Summer of Code — OpenPrinting, The Linux Foundation May 2018 – Aug 2018
Open Source Developer Remote

- Developed a Common Print Dialog backend serving all available print queues (CUPS, Google Cloud Print) over D-Bus.

PUBLICATIONS

Rex: Closing the language-verifier gap with safe and usable kernel extensions. Jinghao Jia, Ruowen Qin, Milo Craun, Egor Lukiyanov, **Ayush Bansal**, Michael V. Le, Hubertus Franke, Hani Jamjoom, Tianyin Xu, Dan Williams. *USENIX Annual Technical Conference (ATC '25)*.

ACADEMIC SERVICE

Artifact Evaluation Committee Aug 2024 – Present

- OSDI '25 · USENIX ATC '25 · SOSP '25 · ACM SIGCOMM '25 · EuroSys '26.

HONORS & AWARDS

NSDI Student Grant Award — travel grant to attend the conference 2025
Academic Excellence Award — exceptional academic performance, 3 consecutive years (IITK) 2020
Exceptional Course Tutor — rated exceptionally by students, AY 2019–20 (IITK) 2019
JEE Advanced — All India Rank 187 out of 200,000 candidates 2016
KVPY Scholarship — All India Rank 52 (IISc & Government of India) 2016
JEE Mains — All India Rank 571 out of 1.5 million candidates 2016
Indian National Chemistry Olympiad — Merit Certificate, National Top 1% 2015
Indian National Astronomy Olympiad — Merit Certificate, National Top 1% 2015

SKILLS

Languages : C/C++, Python, Golang, Scala, Rust, JavaScript/TypeScript, Haskell, Ruby, Java, Verilog
Web : Akka (Scala), Gin (Golang), Node.js, React, Flask, Ruby on Rails, Django
Utilities : Docker, Kubernetes, eBPF, Ansible, AWS, Azure, Kafka, GDB, Postgres, MySQL, MongoDB, LaTeX
Quantum : Qiskit (IBM Quantum), Stim, Munich Quantum Toolkit (MQT), Cirq (Google Quantum AI)